Early Bearish News in a Bubble-Prone Asset Market

EDSL shared-state pilot

EDSL replication study

October 11, 2026

Abstract

Six of twelve traders received a private downside analyst forecast before the first order in a 30-period call market. The remaining market rules and the behavioral instructions matched a separate bullish-news pilot: four momentum buyers, four bubble riders, and four profit-taking sellers. The bearish update did not prevent an early rise. Prices increased from $15.25 to $20.00 by period 10 and reached $26.50 in period 29; 51 shares traded across 22 periods. Treated and untreated price forecasts converged quickly. First-period rationales show that traders often prioritized their assigned behavior over the private downside signal. The result is one exploratory market session, not an estimate of a causal news effect.

1 Market and intervention

The market has 12 traders, each starting with $100 cash and four shares. Cash earns 5% interest; each share pays $0.40 or $1.00 with equal probability; remaining shares are redeemed for $14 after period 30. The deterministic uniform-price call market uses the same settlement and payoff rules as the bullish bubble pilot.

All traders receive one of three behavioral instructions: four momentum buyers, four bubble riders, or four profit-taking sellers. The instructions provide opening order anchors to make two-sided trade more likely. Two participants from each behavior group are assigned to receive the private update. Before period 1, the report says expected resale demand has weakened and prices could fall toward $10 by period 10. The update clearly says this is an uncertain forecast and that dividends and the actual $14 redemption are unchanged. Recipients: trader-00 and trader-01 (momentum), trader-05 and trader-07 (bubble riders), trader-08 and trader-10 (profit takers).

2 Observed results

Measure	Pilot result
Trader decisions	360
Settlements	30 / 30
Private update deliveries	6
Completed workflow items	396 / 396
Shares traded	51
Periods with trade	22 / 30
Peak transaction price	$26.50
Terminal shares	0 for every trader
Terminal cash range	$538.65–$799.35

Transactions occurred in periods 1–4, 6–10, 14–25, and 29. The price was $15.25 in period 1, rose to $20.00 in period 10, and continued to $26.50 in period 29. It exceeded the $14 terminal redemption value throughout observed transactions. The news forecast of a possible fall toward $10 was not realized in this market path.

Across 360 decisions, traders submitted 171 buy orders, 99 sell orders, and 90 holds. Momentum traders submitted 116 buys. Bubble riders submitted 40 buys and 76 sells, consistent with building positions and then offering shares. The profit-taker group chose hold 86 times, sell 19 times, and buy 15 times.

In period 1, update recipients' mean current-price forecast was $14.50, versus $15.00 for the no-news group. By period 10, means were $19.67 and $19.75; by period 25 they were $25.83 and $26.00. Treated momentum traders explicitly said they would buy early despite the update; treated bubble riders emphasized speculative demand; treated profit takers cited the update as a reason to sell. These are descriptive responses from six traders per group in a single shared market.

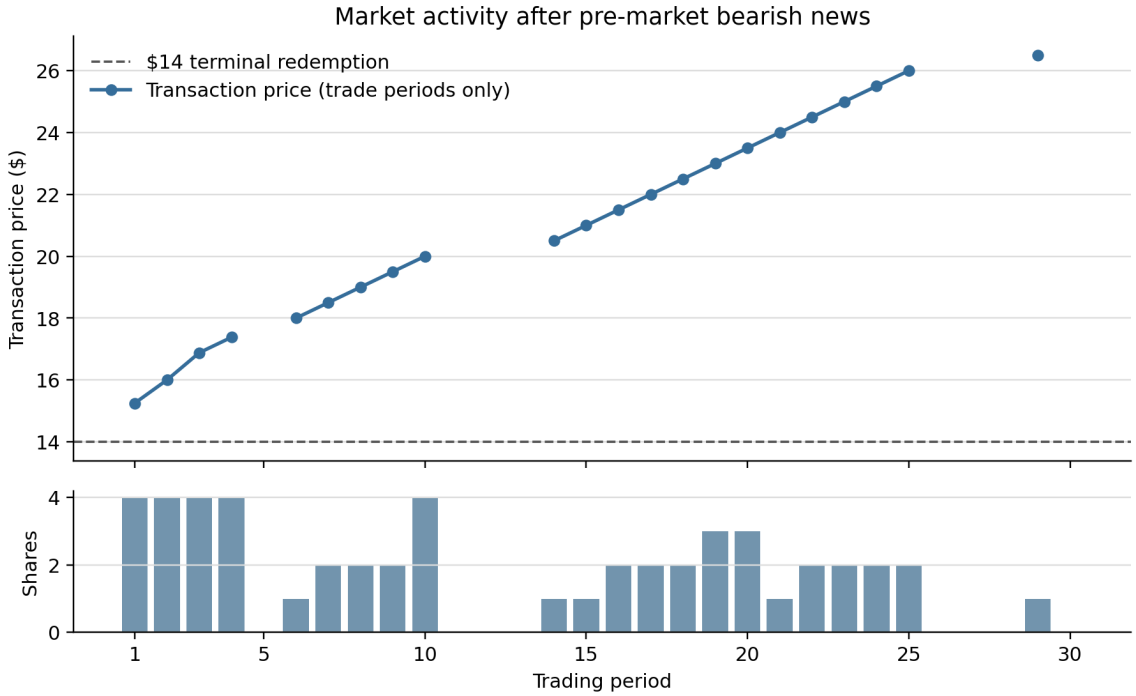


Figure 1: Transaction prices and traded volume. The dashed line marks the $14 terminal redemption amount. Missing prices indicate no transaction.

Relative to the separate bullish-news pilot, this run traded 51 rather than 22 shares and reached $26.50 rather than $18.00. Since each comparison is a single independently run market, differences cannot be attributed to news direction or timing. The pattern is consistent with behavior prompts and market feedback overwhelming a private downside forecast, but repeated markets are needed to test that interpretation.

3 Shared-state machine

The experiment extends EDSL's native `call_market` machine with a map that stores private messages, a command to publish a message, and a participant view that selects the entry for the current trader. The map is omitted from the trader-facing view; each participant receives only their own update.

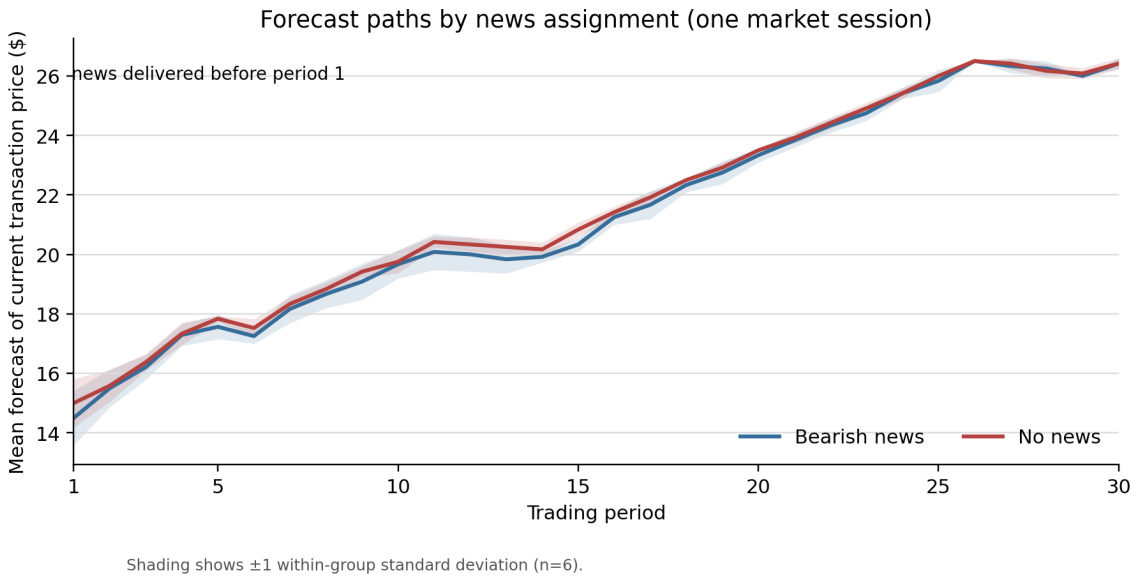


Figure 2: Mean current-price forecasts by news assignment. Shading shows one within-group standard deviation ($n = 6$), not a confidence interval. The update was delivered before period 1.

```
machine = call_market(TRADERS, rules=RULES, seed=SEED)
machine.fields["private_updates"] = state_field(StateType.map(), {})
machine.commands["publish_update"] = Command(
    inputs={"recipient": StateType.text(), "message": StateType.text()},
    effects=(put("private_updates", arg("recipient"), arg("message")),),
)
machine.view["your_update"] = field("private_updates").get(current("name"))
machine.validate()
```

Each write is an exchange task assigned to the scripted executor. Six such tasks are placed before period 1. The first trader order step depends on all six, which guarantees the state write completes before the initial decision. The view expression returns an empty value to traders absent from the private map.

```
for recipient in recipients:
    publish_steps.append(workflow.step(
        f"publish-private-update-{recipient}", update_survey,
        assigned_to=role("exchange"),
        writes=(market.publish_update(
            recipient=recipient, message=UPDATE,
        ),),
    ))
previous = publish_steps
for period in range(1, PERIODS + 1):
    orders = workflow.step(
        f"orders-{period}", Survey([decision]),
        assigned_to=role("trader"), after=previous,
        reads=(market.read(),),
        writes=(market.submit(
            trader=current.agent.name, period=period,
            decision=decision.answer,
        ),),
    )
    settled = workflow.step(
```

```
        f"settle-{period}", settlement_survey,
        assigned_to=role("exchange"), after=orders,
        writes=(market.settle(period=period),),
    )
    previous = settled
```

The first period therefore begins after information publication; later periods retain the ordinary order-then-settlement chain. The six information tasks do not call a model. The detailed machine diagram was generated by EDSL's native `Machine.show_diagram` method.

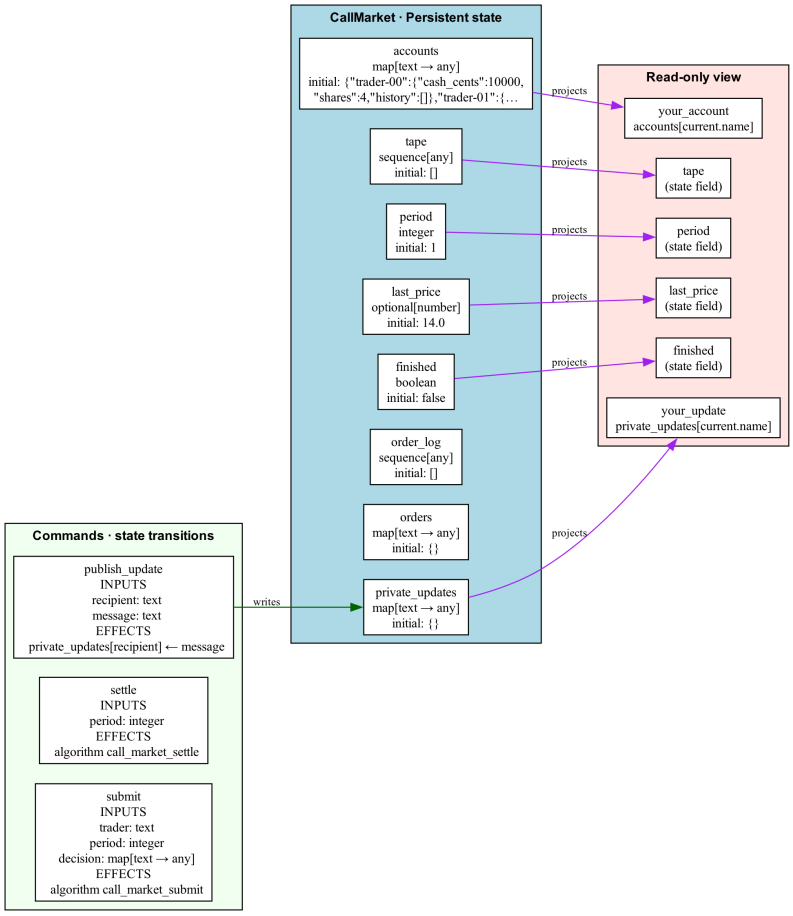


Figure 3: Detailed call-market machine with the private update field and command.

4 Run and limits

The experiment was run locally through EDSL with the configured provider credentials. The workflow, jobs, model calls, shared-state SQLite log, and completion record are preserved in `pilot_01/`. The standalone specification can be rebuilt without model calls using:

```
python3 edsl_replication/studies/asset_market_bearish_early/experiment.py \\
  --output edsl_replication/studies/asset_market_bearish_early/pilot_01 \\
  --prepare-only
```

This one market has a mixed treatment and one shared public tape, so traders affect one another. A factorial design should vary news sign, delivery time, and behavioral prompts independently over

repeated seeds. That would separate the effect of bearish information from assigned speculative behavior and endogenous price feedback.